

State-of-the-Art Construction of IDB Digital Twins for Smart Biomanufacturing

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Introduction of Digital Twin

A digital twin in upstream and downstream bioprocessing is a dynamic, real-time digital representation of the physical bioprocess system. It continuously integrates live data from on-line and at-line sensors, as well as supplementary off-line analytical measurements, to reflect the current state of the physical asset. This comprehensive and synchronized digital profile enables proactive and real-time optimization of bioprocess performance by facilitating advanced monitoring, predictive analysis, and control strategies. While digital models form the foundational basis of digital twins by simulating system behaviors, a digital twin distinguishes itself by being a live, data-driven model that evolves in tandem with the actual process, thereby providing actionable insights throughout the bioprocess lifecycle.

Digital Twin Approach and Model

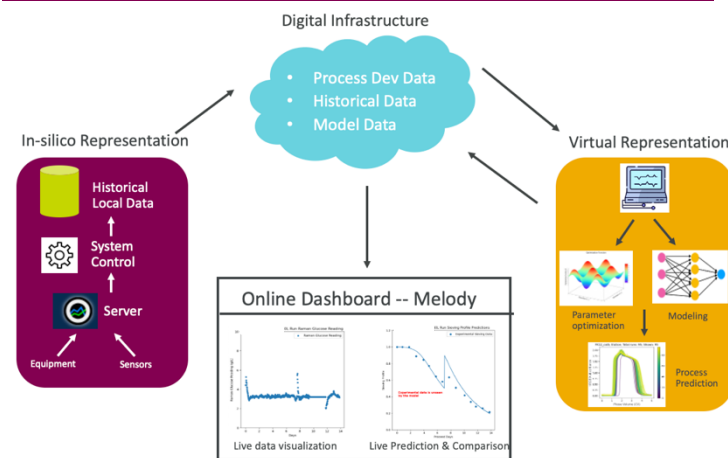


Figure 1. Enabling Digital Twin-Driven Bioprocessing: Foundations and Applications

This flowchart illustrates the digital twin system, where process and model data are collected and processed digitally, generating live virtual models for prediction and optimization. Model-predictive control is the next step after live predictions.

Methods

- All code is written in **Python and Jupyter Notebooks**: enabling rapid prototyping, simulation, and advanced data analytics.
- An interactive front-end dashboard is available on **Melody**, facilitating near real-time visualization and prediction.

IDB Digital Twin Use Cases

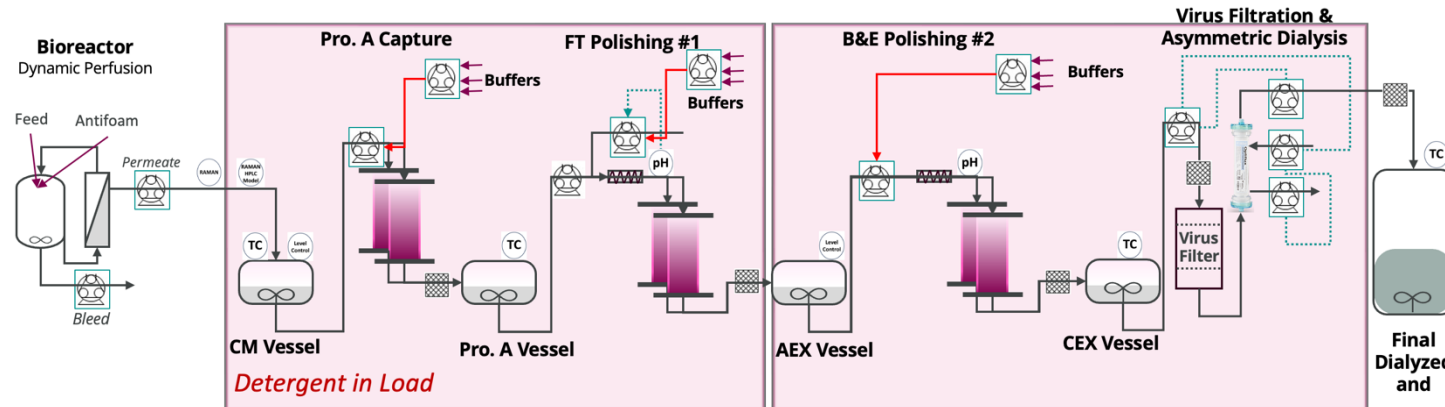


Figure 2. IDB Process Use Cases, by Jon Coffman. TFF Sieving (filter Swap), CM tank titer (proA loading), Chromatograms for error detection.

Results

Upstream IDB Development: Multi-Scale Glucose Uptake – Total Titer Model Fitting

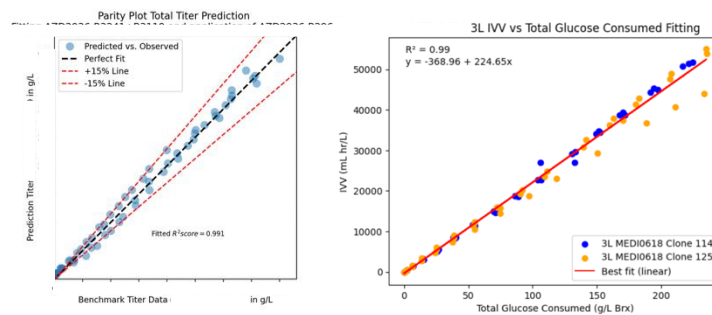


Figure 3. Glucose uptake vs total titer fitting across multi-scale from ambr250, 3L to 50L and 500L for multiple projects.

Upstream IDB Development Model Flow

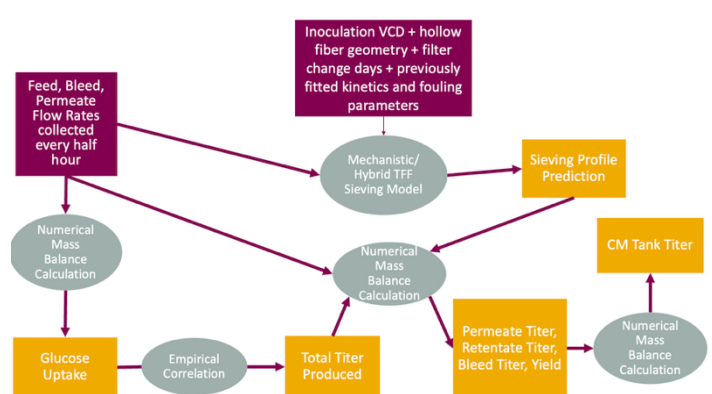


Figure 4. Upstream IDB Prediction Flowchart.

Prediction includes Sieving, Permeate titer, CM tank titer, yield, cumulative permeate/bleed & retentate titer.

Near Real-Time IDB DT Prediction on Melody

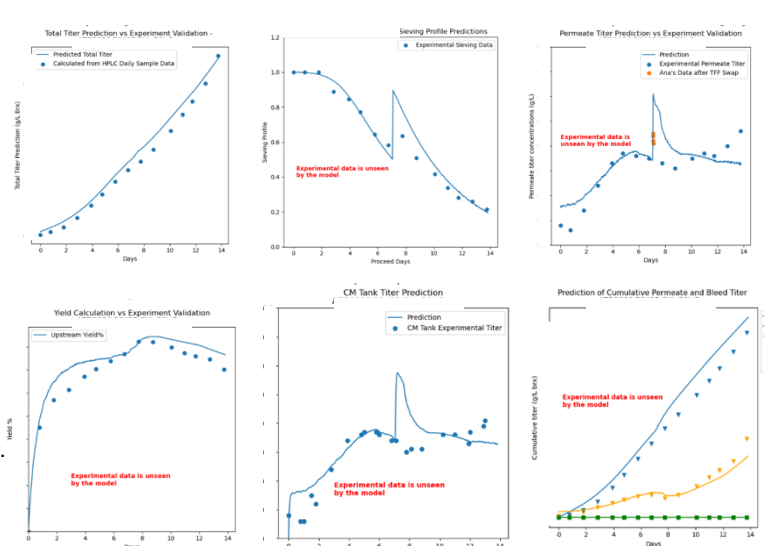


Figure 7. IDB Upstream Digital Twin Melody Example. From left to right: total titer, sieving, permeate titer, yield, CM tank titer, cumulative permeate/bleed & retentate titer prediction for Project B 50L run.

Conclusions

- Facilitates predictive maintenance by continuously monitoring equipment health and performance, leading to **early detection of potential issues and more efficient maintenance scheduling**.
- Reduces unplanned downtime and overall maintenance costs** through proactive identification and resolution of equipment faults.
- Optimizes bioprocesses and operational workflows by integrating **real-time data and advanced analytics** for enhanced performance.
- Enhances safety and reduces operational risks** by providing timely alerts and supporting root cause analysis of deviations.
- Supports improved **decision-making** by delivering actionable, data-driven insights to operators and engineers.
- Enables real-time prediction, **error detection**, and advanced process control using live digital models that reflect the current state of the system.

Acknowledgements

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